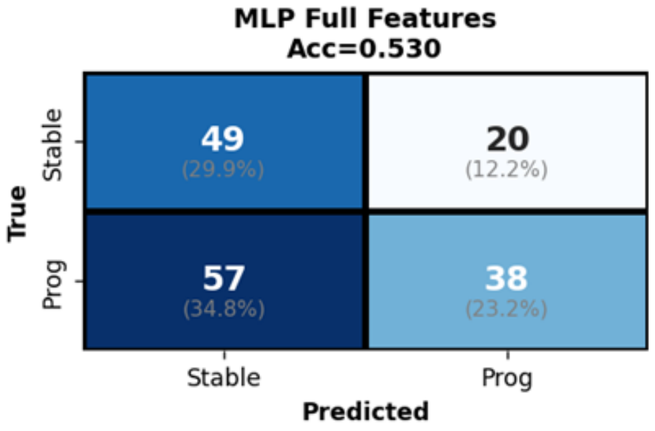
PROGRESS PREDICTION

CNN- slope prediction metrics

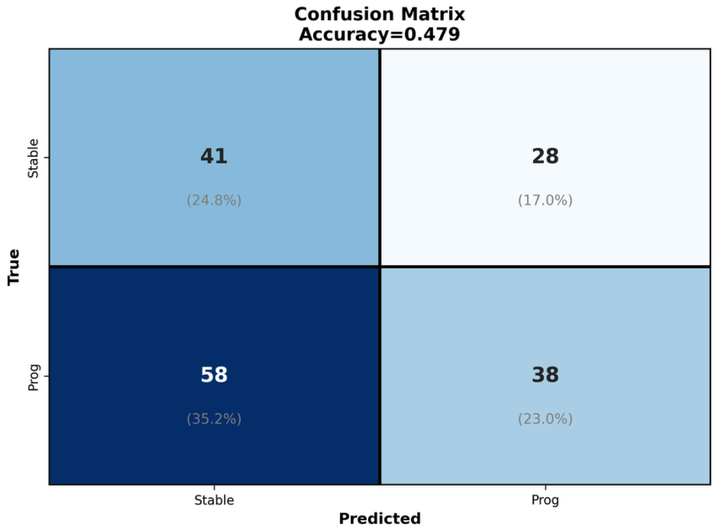
Comparison	Accuracy	Precision	Recall	F1-Score	Specificity	AUC	AP
CNN only	0.505 ± 0.070	0.671 ± 0.135	0.399 ± 0.154	0.464 ± 0.123	0.694 ± 0.196	0.580 ± 0.086	0.653 ± 0.111
CNN+Demographics	0.512 ± 0.078	0.673 ± 0.145	0.406 ± 0.153	0.471 ± 0.134	0.694 ± 0.196	0.581 ± 0.086	0.655 ± 0.117
CNN+Handcrafted	0.526 ± 0.084	0.709 ± 0.180	0.407 ± 0.151	0.480 ± 0.133	0.721 ± 0.210	**0.593 ± 0.085**	**0.667 ± 0.115**
CNN+Demographics+Handcrafted	**0.533 ± 0.093**	**0.716 ± 0.182**	**0.410 ± 0.142**	**0.488 ± 0.135**	**0.739 ± 0.206**	0.585 ± 0.067	0.664 ± 0.107

The CNN-only model still collapes having an average standard deviation between folds of 0.922.

Adding the other features helps the model avoiding the collapse increasing the stardard deviation.



FVC Predictor



First approach:

- Direct prediction of FVC value at week 52

But also here model collapses towards the mean

Method	Accuracy	Precision	Recall	F1	Specificity	AUC	AP
Direct	0.478 ± 0.073	0.550 ± 0.191	0.386 ± 0.141	0.448 ± 0.157	0.597 ± 0.150	0.483 ± 0.115	0.582 ± 0.129

To avoid model collapsing:

- Tried dual head
  - One predicted FVC baseline from Ct scans
  - One predicted the Delta FVC

But results where the same

One (big) problem - Baseline FVC (Intercept)

Once the slope has been predicted by the CNN, to predict the FVC:

FVC(52) = predicted_slope * 52 + intercept_0

The intercept at week 0 is a product of the regression initially applied on the FVC data to extract the slope (used as target in the CNN).

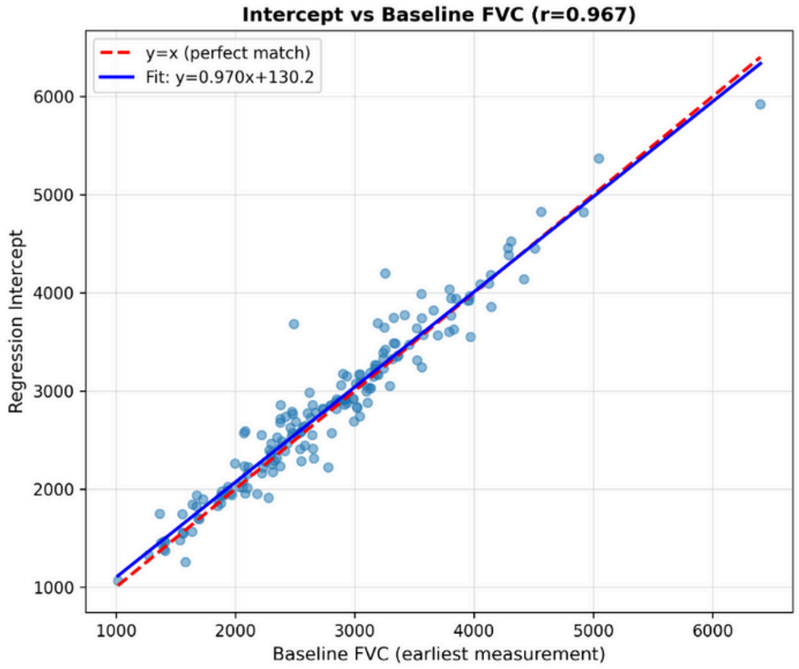
But we shouldn't have this in Test inference..... as the main input is the CT SCAN.

So when testing:

FVC(52) = predicted_slope * 52 + ?????

Therefore the options I thought were:

- CNN predict slope and intercept at week 0 .... (Hard)
- Use Intercept at week 0 as input other than the CT SCAN , and use it like a Baseline FVC but this would introduce data leakage
- Use Baseline FVC of patients ..... but there is the usual problem of missing data (90%)
    - But if add a a tolerance [-5,+10] → 52.2% of patie



RECAP ON FVC(0):

- 18/176 - 10% of patients have week 0 value
- The rest is sparse, reaching also week 79....
- The MAE with the intercept is 90 ml
    - Better than taking first possible value

Final option:

- CNN shouldn't predict intercept, but the state of the patient at the CT SCAN

Possible model:

1. CT → Z (state of the lungs)
2. Two heads
    a. Baseline : Z → FVC(0)
    b. Slope : Z → Slope
3. Reconstruction → FVC(t) = FVC(0) + Slope * t → FVC(52)
4. Loss → MSE(True_FVC52, Predicted_FVC52) + lambda * MSE(True_slope,Predicted_slope)
   Therefore No Loss on the baseline (which we don't have)

Therefore the 3 approaches in the end would be:

1) FVC@52 :
Feature embedding + handcrafted + demographics → FVC(52) ✓

2) FVC@52 - slope:
Feature embedding predict FVC(0) and slope, then reconstruct FVC(52)

2a) Use Demographics to refine slope

2b) Use Handcrafted to refine slope

2c) Use both to refine slope

3) Progression prediction:
Feature embedding + handcrafted + demographics → Stable/Progressed